

Aarya Birwadkar

Batch C2 PRN-1262252502

Data Structures Assignment 7

Code:

```
#include<stdio.h>
#include<ctype.h>
#define MAX 100
int stack[MAX];
int top=-1;
void push(int value)
{
    stack[++top]=value;
}
int pop()
{
    return stack[top-1];
}
int evaluatepostfix(char postfix[])
{
    int i,a,b,result;
    for(i=0;postfix[i]!='\0';i++)
    {
        if(isdigit(postfix[i]))
        {
            push(postfix[i]-'0');
        }
        else
        {
            {
```

```
b=pop();
a=pop();
switch(postfix[i])
{
case '+':
result=a+b;
break;
case '-':
result=a-b;
break;
case '*':
result=a*b;
break;
case '/':
result=a/b;
break;
case '%':
result=a%b;
break;
default:
printf("invalid operation");
}
push(result);
}
}
return pop();
}
int main()
{
```

```
char postfix[MAX];  
int result;  
printf("enter postfix expression:");  
scanf("%s",postfix);  
result=evaluatepostfix(postfix);  
printf("result=%d\n",result);  
return 0;  
}
```

Output:

```
vyaas307@VY029-26 ~ % ./a.out  
enter postfix expression:24/56*-  
result=25
```